

Theory Answers

This handbook contains definitions, equations, examples, and simple text diagrams for each topic.

Part A – Theory (Short Questions)

1. Define **Mean, Median, Mode** with an example in the context of employee salaries.
2. Differentiate between **Range and Variance** with examples.
3. What is the difference between **Normal Distribution** and **Poisson Distribution**?
4. Explain **Skewness** and give one workplace-related example.
5. Define **Conditional Probability** and explain how it applies in promotions.
6. Explain **Independent vs Mutually Exclusive Events** with examples.
7. What does **Bayes Theorem** help us in real-world decision-making?
8. Write a short note on **PCA (Principal Component Analysis)** in simple words.

1. Mean, Median and Mode

Mean: Average of all observations.

Equation: Mean = $\Sigma x / n$.

Median: Middle value after sorting. Odd: $(n+1)/2$ th observation. Even: average of two middle values.

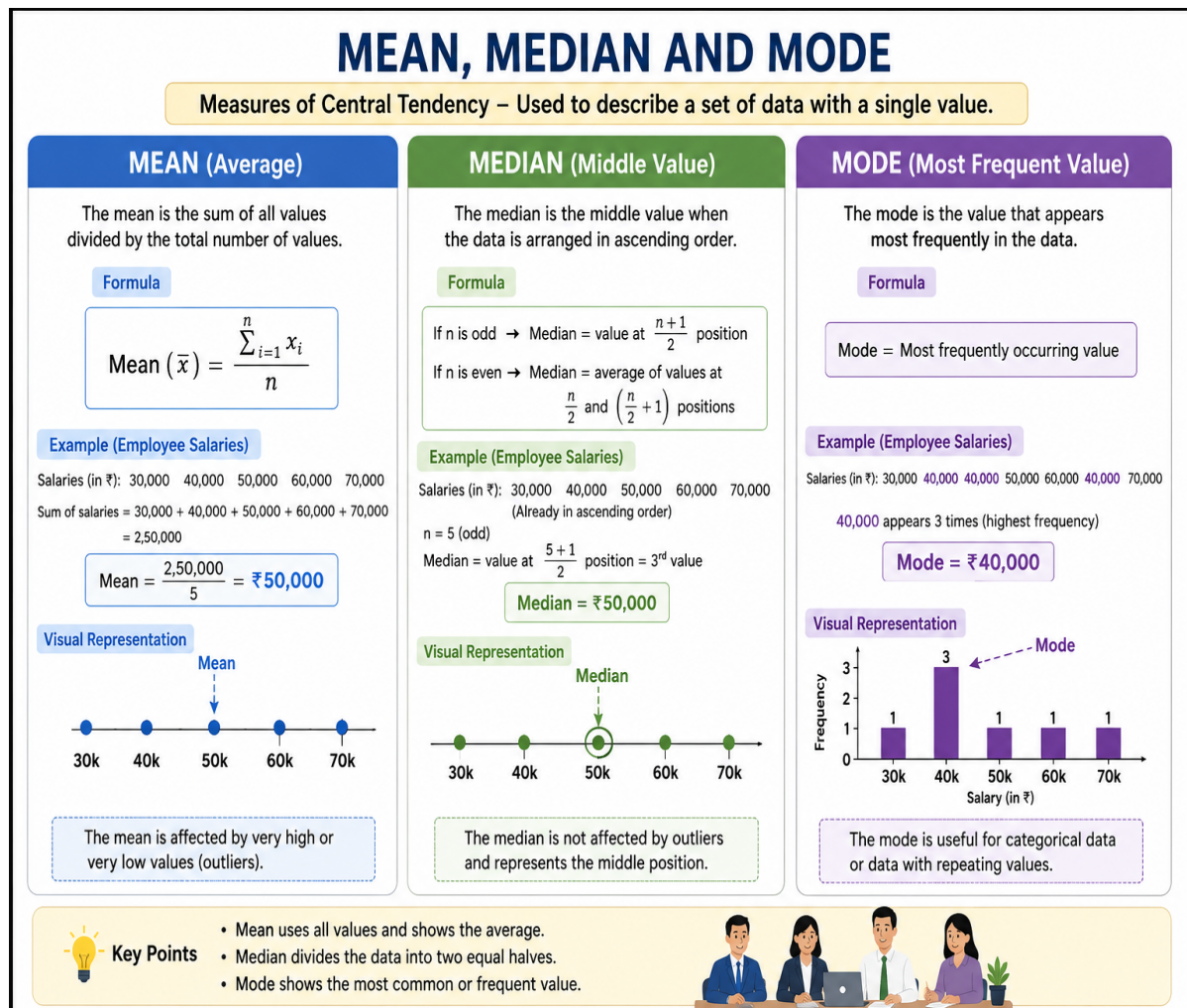
Mode: Most frequently occurring value.

Example salaries: 30000, 35000, 35000, 40000, 50000.

Mean=38000, Median=35000, Mode=35000.

Diagram:

30000 → 35000 → **35000** → 40000 → 50000



2. Range and Variance

Range measures spread using only minimum and maximum values.

Equation: Range = $X_{\max} - X_{\min}$.

Variance measures overall spread around the mean.

Population: $\sigma^2 = \Sigma(x - \mu)^2 / N$.

Sample: $s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$.

Diagram: Low spread: ●●●● High spread: ●●●●●

RANGE AND VARIANCE

Measures of Dispersion – Used to measure the spread or variability of data.

RANGE

Range is the difference between the largest value and the smallest value in a dataset.

Formula

$$\text{Range} = X_{\max} - X_{\min}$$

$X_{\max}$ = Maximum value
 $X_{\min}$ = Minimum value

Example (Employee Salaries)

Salaries (in ₹): 30,000 40,000 50,000 60,000 70,000 80,000

Minimum Maximum

Range = $80,000 - 30,000 = ₹50,000$

Visual Representation

Key Point: Range is easy to calculate but is affected by extreme values (outliers).

VARIANCE

Variance measures how far the data values are spread out from the mean (average).

Formulas

Population Variance

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

Sample Variance

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Where,
 x_i = each value $\bar{x}$ = sample mean
 μ = population mean N = population size, n = sample size

Example (Employee Salaries)

Salaries (in ₹): 30,000 40,000 50,000 60,000 70,000

Mean ($\bar{x}$) = $\frac{30,000 + 40,000 + 50,000 + 60,000 + 70,000}{5} = 50,000$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
30,000	-20,000	400,000,000
40,000	-10,000	100,000,000
50,000	0	0
60,000	10,000	100,000,000
70,000	20,000	400,000,000
Total		1,000,000,000

Population Variance,

$$\sigma^2 = \frac{1,000,000,000}{5} = 200,000,000$$

Sample Variance,

$$s^2 = \frac{1,000,000,000}{5 - 1} = 250,000,000$$

Key Point: Variance considers all values in the dataset and is not affected as much by extreme values as range.

RANGE vs VARIANCE

Range uses only two values (max and min).	Variance uses all values.	Range is simple but less reliable.	Variance is more reliable and stable.	Variance is widely used in statistics and machine learning.
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3. Normal vs Poisson Distribution

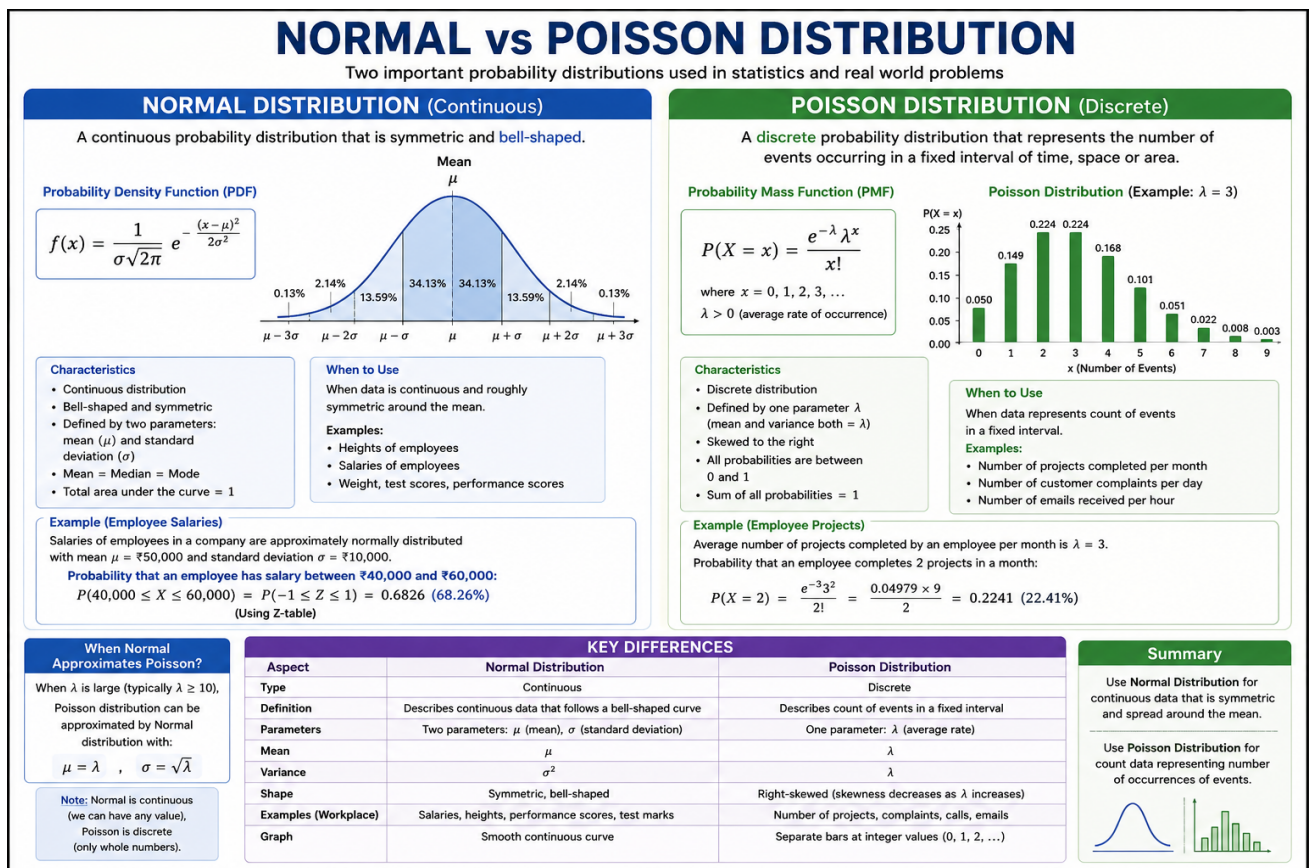
Normal Distribution: Continuous, symmetric bell curve.

Equation: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.

Poisson Distribution: Discrete count distribution.

Equation: $P(X=x) = e^{-\lambda} \frac{\lambda^x}{x!}$.

Diagram: Bell curve:  ; Poisson: 



4. Skewness

Skewness measures asymmetry.

Equation: Skewness = $3(\text{Mean} - \text{Median}) / \sigma$.

Positive skew: tail right. **Negative skew:** tail left.

SKEWNESS

Skewness measures the asymmetry of a distribution (how much the data is tilted to one side).

1. DEFINITION

Skewness indicates the direction and degree of asymmetry in a dataset.

If the distribution is perfectly symmetric, skewness = 0.

3. TYPES OF SKEWNESS

(a) Positive Skew
(Right Skewed)

Mode < Median < Mean

Long tail on the right side.
Mean > Median > Mode
Skewness > 0

(b) Negative Skew
(Left Skewed)

Mean < Median < Mode

Long tail on the left side.
Mean < Median < Mode
Skewness < 0

(c) Zero Skew
(Symmetric)

Mean = Median = Mode

Perfectly symmetric.
Mean < Median = Mode
Skewness = 0

4. INTERPRETATION

Skewness (Sk)	Interpretation	Shape
Sk > 0	Positive Skew (Right Skewed)	
Sk < 0	Negative Skew (Left Skewed)	
Sk = 0	Zero Skew (Symmetric)	

Key Point
The farther the skewness value from 0, the more asymmetric the data.

5. EXAMPLES (Workplace Related)

(a) Positive Skew Example

Most employees earn moderate salaries, but a few earn very high salaries.

Salaries (in ₹): 25,000 30,000 35,000 40,000 45,000 120,000

Step 1: Calculate Mean, Median, SD

Mean ($\bar{x}$) = ₹49,167

Median = ₹37,500

Standard Deviation (σ) = ₹30,200 (approx.)

Step 2: Apply Formula

$$Sk = \frac{3(\bar{x} - \text{Median})}{\sigma}$$

$$= \frac{3(49,167 - 37,500)}{30,200}$$

$$= \frac{35,001}{30,200} = 1.16$$

Since Sk > 0, the distribution is **Positively Skewed** (right skewed).
A few very high salaries pull the mean to the right.

(b) Negative Skew Example

Most employees earn high salaries, but a few earn very low salaries.

Salaries (in ₹): 15,000 60,000 70,000 75,000 80,000 95,000

Step 1: Calculate Mean, Median, SD

Mean ($\bar{x}$) = ₹65,833

Median = ₹72,500

Standard Deviation (σ) = ₹25,300 (approx.)

Step 2: Apply Formula

$$Sk = \frac{3(\bar{x} - \text{Median})}{\sigma}$$

$$= \frac{3(65,833 - 72,500)}{25,300}$$

$$= \frac{-20,001}{25,200} = -0.79$$

Since Sk < 0, the distribution is **Negatively Skewed** (left skewed).
A few very low salaries pull the mean to the left.

(c) Zero Skew Example

Salaries are symmetrically distributed.

Salaries (in ₹): 40,000 50,000 60,000 70,000 80,000

Step 1: Calculate Mean, Median, SD

Mean ($\bar{x}$) = ₹60,000

Median = ₹60,000

Standard Deviation (σ) = ₹15,800 (approx.)

Step 2: Apply Formula

$$Sk = \frac{3(\bar{x} - \text{Median})}{\sigma}$$

$$= \frac{3(60,000 - 60,000)}{15,800}$$

$$= \frac{0}{15,800} = 0$$

Since Sk = 0, the distribution is **Symmetric** (no skewness).
Mean, Median and Mode are equal.

6. IMPORTANCE OF SKEWNESS

- Helps in understanding the shape of data.
- Useful in decision making and planning.
- Helps HR / Management to understand salary distribution, performance scores, etc.

7. RULE OF THUMB

- $|Sk| < 0.5 \rightarrow$ Nearly Symmetric
- $0.5 \leq |Sk| < 1 \rightarrow$ Moderately Skewed
- $|Sk| \geq 1 \rightarrow$ Highly Skewed

5. Conditional Probability

Probability of A given B.

Equation: $P(A|B)=P(A\cap B)/P(B)$.

Example: Probability(Employee promoted | Score>80).

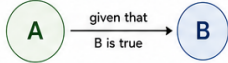
CONDITIONAL PROBABILITY

The probability of an event occurring given that another event has already occurred.

1. DEFINITION

Conditional Probability is the probability of event A occurring given that event B has already occurred.

In Words
 $P(A|B)$ means "Probability of A, given that B is true".



2. FORMULA

The formula for conditional probability is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Where,

- $P(A|B)$: Probability of A given B
- $P(A \cap B)$: Probability that both A and B occur
- $P(B)$: Probability of B occurring

Note: $P(B) > 0$ (B must have a non-zero probability)

3. EXAMPLE (Workplace Scenario)

A company has 200 employees.

- 120 employees have performance score > 80.
- 30 employees have performance score > 80 and got promoted.

Event Definitions:

A : Employee gets promoted
B : Employee has performance score > 80

Find: Probability that an employee gets promoted given that their performance score is > 80 $\rightarrow P(A|B)$

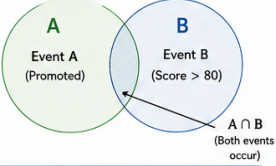
	Promoted (A)	Not Promoted (A ^c)	Total
Performance > 80 (B)	30	90	120
Performance ≤ 80 (B ^c)	20	60	80
Total	50	150	200

Solution:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{30/200}{120/200} = \frac{30}{120} = 0.25$$

Answer: The probability that an employee gets promoted given that their performance score is > 80 is 0.25 or 25%.

4. HOW IT WORKS (Venn Diagram)



$$P(A|B) = \frac{\text{Area of } (A \cap B)}{\text{Area of } (B)}$$

We only consider the part of A that falls inside B, since B is already known to have occurred.

5. MORE EXAMPLES

(i) **Quality Check Example**
Out of 500 products, 80 are defective.
Out of the defective products, 20 are returned by customers.
Find: Probability that a product is returned given that it is defective.

$$P(\text{Returned}|\text{Defective}) = \frac{20}{80} = 0.25 \text{ (25\%)}$$

(ii) **Training Example**
Out of 150 employees, 60 attended training program.
Out of these 60, 45 improved their performance.
Find: Probability that an employee improved performance given that they attended training.

$$P(\text{Improved}|\text{Attended}) = \frac{45}{60} = 0.75 \text{ (75\%)}$$

6. APPLICATIONS IN WORKPLACE

**Recruitment:** Probability of selecting a candidate given that they passed the test.

**Performance Analysis:** Probability of promotion given that the employee met the target.

**Risk Assessment:** Probability of an incident given that a certain condition occurred.

**Customer Behavior:** Probability of a customer buying a product given that they viewed the ad.

**KEY TAKEAWAY**
Conditional probability helps us make better decisions by updating probabilities based on additional information.

Important Properties

- $P(A|B) = 0$ if $P(A \cap B) = 0$
- $P(A|B) = 1$ if $A \subseteq B$ (A always occurs when B occurs)
- $P(A|B) + P(A^c|B) = 1$

Numerical Check

From the example in section 3:

$$P(\text{Not Promoted}|\text{Score} > 80) = \frac{90}{120} = 0.75$$

Check: $P(\text{Promoted}|B) + P(\text{Not Promoted}|B) = 0.25 + 0.75 = 1 \checkmark$

Remember!
Always check the condition (first event) before evaluating the probability of the second event. 

6. Independent vs Mutually Exclusive

Independent: $P(A \cap B) = P(A)P(B)$.

Mutually Exclusive: $P(A \cap B) = 0$.

Example: Promotion Yes and Promotion No are mutually exclusive.

INDEPENDENT vs MUTUALLY EXCLUSIVE EVENTS

Two important types of events in probability theory.

1. INDEPENDENT EVENTS

Two events are **independent** if the occurrence of one event does not affect the probability of the other event.

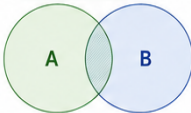
Formula

$$P(A \cap B) = P(A) \times P(B)$$

or

$$P(A | B) = P(A) \text{ and } P(B | A) = P(B)$$

Venn Diagram

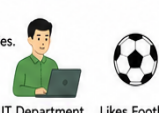


A and B can occur at the same time.

Example (Workplace Related)

A company has employees from different departments and with different hobbies.

- A: Employee works in the IT department.
- B: Employee likes playing football.



Explanation:
Whether an employee works in IT or not does not affect whether they like football or not. So, events A and B are independent.

Assume: $P(A) = 0.40$ (40% employees are in IT)
 $P(B) = 0.30$ (30% employees like football)

Then, $P(A \cap B) = P(A) \times P(B) = 0.40 \times 0.30 = 0.12$
(12% employees are in IT and like football)

2. MUTUALLY EXCLUSIVE EVENTS

Two events are **mutually exclusive** if they cannot occur at the same time.

Formula

$$P(A \cap B) = 0$$

or

$$P(A | B) = 0 \text{ and } P(B | A) = 0$$

Venn Diagram



A and B cannot occur at the same time.

Example (Workplace Related)

Consider the promotion status of an employee.

- A: Employee gets promoted this year.
- B: Employee does not get promoted this year.



Explanation:
An employee cannot be both promoted and not promoted at the same time. So, events A and B are mutually exclusive.

Assume: $P(A) = 0.25$ (25% employees get promoted)
 $P(B) = 0.75$ (75% employees do not get promoted)

Then, $P(A \cap B) = 0$
(Because both events cannot occur together)

3. KEY DIFFERENCES

Aspect	Independent Events	Mutually Exclusive Events
Definition	Occurrence of one event does not affect the other.	Both events cannot occur at the same time.
Formula	$P(A \cap B) = P(A) \times P(B)$	$P(A \cap B) = 0$
Venn Diagram	 Overlap exists.	 No overlap.
Example (Workplace)	Working in IT department and liking football.	Employee gets promoted and employee does not get promoted.
Conditional Probability	$P(A B) = P(A)$	$P(A B) = 0$

4. IMPORTANT NOTES

- ✓ All mutually exclusive events are independent only when at most one of them can occur.
- ✓ Two events can be independent but not mutually exclusive.
- ✓ If A and B are independent, it does not mean $P(A \cap B) = 0$.
- ✓ Mutually exclusive events have no common outcomes.

5. QUICK SUMMARY

✓ **Independent Events**
Can occur together or separately.

$$P(A \cap B) = P(A) \times P(B)$$

✗ **Mutually Exclusive Events**
Cannot occur together.

$$P(A \cap B) = 0$$

7. Bayes' Theorem

Updates probability using new evidence.

Equation: $P(A|B)=P(B|A)P(A)/P(B)$.

Used to estimate promotion probability after observing high performance.

BAYES' THEOREM

A way to update probabilities based on new evidence.

1. DEFINITION

Bayes' Theorem relates the conditional and marginal probabilities of random events.

BAYES' THEOREM FORMULA

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Where,

- $P(A|B)$: Posterior probability (Probability of A given B)
- $P(B|A)$: Likelihood (Probability of B given A)
- $P(A)$: Prior probability (Initial probability of A)
- $P(B)$: Evidence or Marginal probability of B

LAW OF TOTAL PROBABILITY

$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i)$$

(Used to calculate $P(B)$ when A has multiple possibilities)

2. INTUITION (IN SIMPLE WORDS)

Bayes' Theorem helps us revise our belief about an event (A) when we get new evidence (B).

Example in words:
You may have an initial belief that an employee is likely to be high performing (Prior). After seeing their project report (Evidence), you update your belief about their performance (Posterior).

3. WORKPLACE EXAMPLE (DIAGNOSING HIGH PERFORMANCE)

A company wants to identify high performing employees using a Performance Test.

- 20% of employees are actually High Performers (H).
- The test correctly identifies 90% of high performers. $P(+|H) = 0.90$
- The test incorrectly marks 15% of low performers as high. $P(+|L) = 0.15$

(Here, "+" means the test result is Positive for high performance.)

Category	Meaning	Probability
$P(H)$	Prior: Employee is a High Performer	0.20
$P(L)$	Prior: Employee is a Low Performer	0.80
$P(+ H)$	Likelihood: Test Positive given High Performer	0.90
$P(+ L)$	Likelihood: Test Positive given Low Performer	0.15

Step 1: Find $P(+)$ using Law of Total Probability

$$P(+) = P(+|H)P(H) + P(+|L)P(L)$$
$$= (0.90)(0.20) + (0.15)(0.80) = 0.18 + 0.12 = 0.30$$

Step 2: Apply Bayes' Theorem

$$P(H|+) = \frac{P(+|H)P(H)}{P(+)} = \frac{(0.90)(0.20)}{0.30} = 0.60$$

Answer: Given that the test result is positive, the probability that the employee is a high performer is **60%**.

4. VISUAL REPRESENTATION

For Two Events (A and B)

Updating Belief

Bayes' Theorem helps this update

5. MORE GENERAL FORM (MULTIPLE HYPOTHESES)

If there are n mutually exclusive and exhaustive hypotheses $H_1, H_2, \dots, H_n$, then

$$P(H_i|E) = \frac{P(E|H_i)P(H_i)}{\sum_{j=1}^n P(E|H_j)P(H_j)}$$

Where:

- H_i : i -th hypothesis
- E : Evidence
- $P(H_i)$: Prior of hypothesis H_i
- $P(E|H_i)$: Likelihood of evidence E given H_i

6. ANOTHER WORKPLACE EXAMPLE

A company recruits from three sources.

- 40% from Referrals (R), 35% from Job Portals (J), 25% from Campus (C).
- Probability that a hired employee performs well (E) from each source:
 - Referrals: 0.70
 - Job Portals: 0.40
 - Campus: 0.50

An employee is hired and performs well. What is the probability that the hire came from Referrals?

Priors

$$P(R) = 0.40$$
$$P(J) = 0.35$$
$$P(C) = 0.25$$

Likelihoods

$$P(E|R) = 0.70$$
$$P(E|J) = 0.40$$
$$P(E|C) = 0.50$$

Using Bayes' Theorem

$$P(R|E) = \frac{P(E|R)P(R)}{P(E)}$$

First, $P(E) = (0.70)(0.40) + (0.40)(0.35) + (0.50)(0.25)$
 $= 0.28 + 0.14 + 0.125 = 0.545$

Therefore, $P(R|E) = \frac{0.70 \times 0.40}{0.545} = 0.514$
So, there is 51.4% chance the hire is from Referrals.

7. KEY POINTS

- ✓ Bayes' Theorem updates our belief about a hypothesis using new evidence.
- ✓ It combines Prior, Likelihood, and Evidence to give Posterior probability.
- ✓ Widely used in HR analytics, medical diagnosis, spam filtering, fraud detection, quality control, decision making, and machine learning.
- ✓ It becomes more accurate as more evidence is available.

8. COMMON APPLICATIONS IN WORKPLACE

- Employee performance prediction
- Recruitment quality evaluation
- Attrition risk assessment
- Customer satisfaction analysis
- Fraud detection
- Product defect prediction

9. SUMMARY

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

In Words

Posterior = $\frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$

Remember: Prior is what you believe before seeing the evidence, Posterior is what you believe after seeing the evidence.

8. Principal Component Analysis (PCA)

PCA reduces many correlated variables into fewer principal components.

Covariance Matrix: $C = (1/(n-1)) \times \sum (X_i - \bar{X})(X_i - \bar{X})^T$.

Eigenvalue Equation: $Av = \lambda v$.

Steps: Standardize → Covariance Matrix → Eigenvalues/Eigenvectors → Select Components → Transform Data.

